

# HOW ENGINEERS BUILT THE TALLEST MARVEL ON DESERT SAND



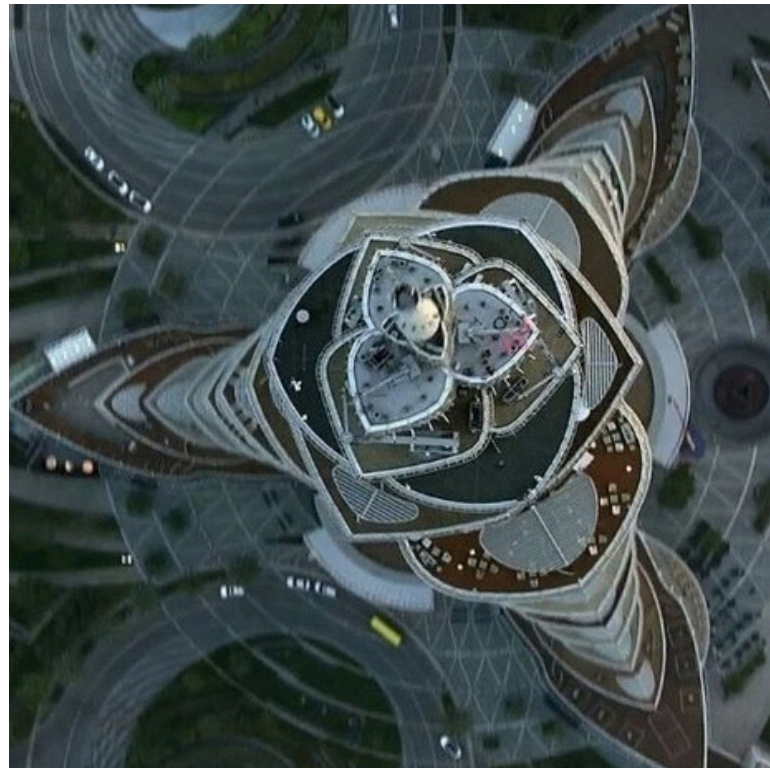
**“Standing 828 meters above Dubai’s sands, the Burj Khalifa turned an impossible vision into engineering reality...!”**

**Location:** Dubai, United Arab Emirates

**Completed:** 2010



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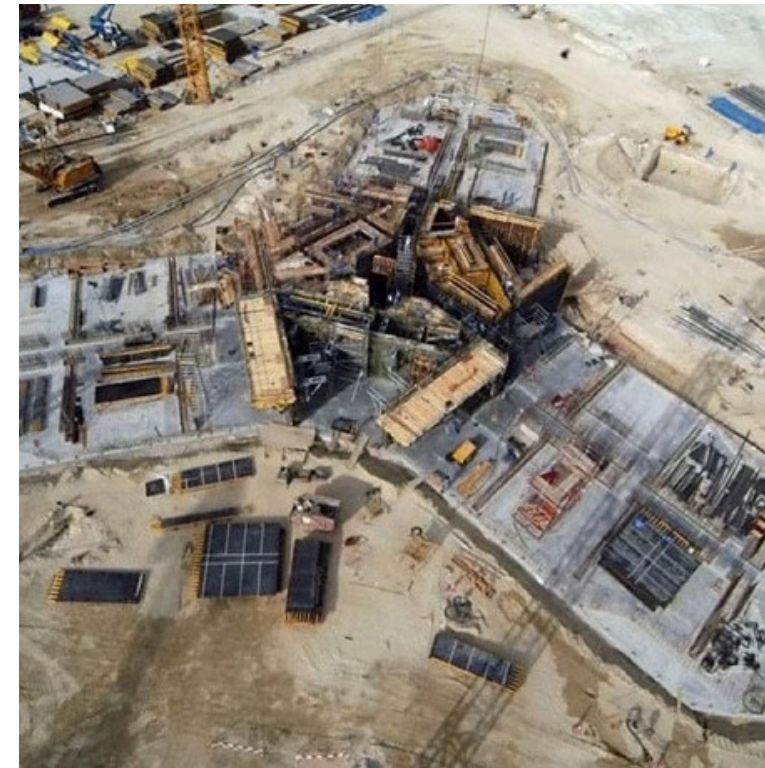
## THE Y - SHAPE

The building is shaped like the letter Y. As it goes up, the steps get smaller. This confuses the strong desert wind so it cannot push the building over



## STRONG CORE

In the middle, there is a strong 6-sided core. It holds the building tight so it does not twist in storms



## THE FOUNDATION

Dubai has sand. not solid rock! So engineers used a 3.7 meter thick concrete base resting on 192 long pillars driven deep into the sand. The building stays up using friction!



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## GLASS WINDOWS

Dubai is very hot. The building is covered with 26000 special glass panels that reflect the sun's heat



## CONCRETE RECORD

Engineers created special liquid concrete and pumped it up 606 meters high. This was a world record



## THE STEEL TOP

The huge steel top weighs 4000 tons. Workers built it inside the top floors and pushed it up using giant hydraulic jacks!

